

HBeethomahler Report

Number of words: 1676

Declaration of AI: The of QuillBot for Grammar and Spell-checking

Contents

1	Overview	3
2	Forensic Hygiene	3
2.1	Evidence Handoff and Preparation	3
2.2	Workstation Integrity and Standards	4
2.3	Verification and Compliance	6
3	Medium	8
3.1	Partition and File System Analysis	8
3.2	System Configuration and Anomalies	9
3.3	Security, Obfuscation and Content Logic	9
4	Operating System	12
4.1	OS Profiling and Enviornment	12
4.2	User Attribution and Security	13
4.3	External Connectivity and Connectivity History . .	14
5	Conclusion	17

1 Overview

This report presents a forensic examination of a digital forensic image (DFI) supplied by West Midlands Police in regards to Henry Beethomahler. The investigation was carried out in strict adherence to the digital forensics standards presented by ACPO in order to ensure objectivity, integrity and admissibility. The methodology employed during this case aligns clearly with ACPO principle 1: "No action taken by law enforcement agencies, persons employed within those agencies or their agents should change data which may subsequently be relied upon in court." All analysis of this case was carried out on a verified forensic copy which prevents modification of the source. In this report I will explore three core areas, forensic hygiene, identification and understanding of the medium and lastly an analysis of the operating systems used.

2 Forensic Hygiene

2.1 Evidence Handoff and Preparation

I was provided with the HBeethomahler Digital Forensic Image by the module leader Harjinder Laille. A DFI is a bit-by-bit copy of a physical storage device which is accurately able to represent all files, folders, unallocated, free and slack space on an operating system. It is worthy to note the differences between the raw data presented and the pre-ingested log files of the case. The raw data is the unprocessed forensic image of the case and are direct copies of storage media. The pre-ingested log files refers to the initial, automatic processing of a data source (such as a disk image, logical file, or forensic image) to identify, parse, and analyse files immediately after they are added to a case. The settings used to the ingest these files are represented in a table below:

Module Name	Module Version
Recent Activity	4.19.3
Hash Lookup	4.19.3
File Type Identification	4.19.3
Extension Mismatch Detector	4.19.3
Embedded File Extractor	4.19.3
Picture Analyzer	4.19.3
Keyword Search	4.19.3
Email Parser	4.19.3
Encryption Detection	4.19.3
Interesting Files Identifier	4.19.3
Central Repository	4.19.3
PhotoRec Carver	7.0
Virtual Machine Extractor	4.19.3
Data Source Integrity	4.19.3

Table 1: Module Names and Versions

2.2 Workstation Integrity and Standards

To ensure a "forensically clean" workstation I allowed Windows Defender to carry out a malware scan on the provided sources. I did this by selecting custom scan and choosing the file that held the HBeethomahler case, the result of this scan was negative. A further precaution to protect my workstation was to disconnect the virtual machine from the network, isolating the host prevents the VM from accessing external networks directly. By doing this a more secure environment is created as it will prevent potential malware on the system from spreading outside of the container of the original VM and into the host system or other VMs.

Run a quick, full, custom, or Windows Defender
Offline scan.

No current threats.

Last scan: 1/28/2026 12:58 PM (custom scan)

0 threats found.

Scan lasted 2 seconds

760 files scanned.

Figure 1: Windows Defender Scan Results

The virtual machine that is being used to examine the HBeethomaler uses a Windows operating system. Below I have provided a table of information about the details of the OS:

Host Name	WINDOWS10-VM
OS Name	Microsoft Windows 10 Enterprise
OS Version	10.0.18363 N/A Build 18363
OS Manufacturer	Microsoft Corporation
OS Configuration	Standalone Workstation
OS Build Type	Multiprocessor Free
Registered Owner	Windows User
Registered Organization	
Product ID	00329-10130-31990-AA053
Original Install Date	8/22/2023, 7:10:46 PM
System Boot Time	2/3/2026, 11:17:48 AM

```

C:\Users\vbouser>systeminfo

Host Name:                WINDOWS10-VM
OS Name:                  Microsoft Windows 10 Enterprise
OS Version:               10.0.18363 N/A Build 18363
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         Windows User
Registered Organization:
Product ID:                00329-10130-31990-AA053
Original Install Date:     8/22/2023, 7:10:46 PM
System Boot Time:          2/3/2026, 11:17:48 AM
System Manufacturer:       innotek GmbH
System Model:              VirtualBox
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: AMD64 Family 23 Model 8 Stepping 2 AuthenticAMD ~3583 Mhz
BIOS Version:              innotek GmbH VirtualBox, 12/1/2006
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+00:00) Dublin, Edinburgh, Lisbon, London
Total Physical Memory:     14,131 MB
Available Physical Memory: 9,555 MB
Virtual Memory: Max Size:  16,307 MB
Virtual Memory: Available: 9,902 MB
Virtual Memory: In Use:    6,405 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\WINDOWS10-VM
Hotfix(s):                 7 Hotfix(s) Installed.
                           [01]: KB5013627
                           [02]: KB4512661
                           [03]: KB4516115
                           [04]: KB4517245
                           [05]: KB4521863
                           [06]: KB5007114
                           [07]: KB4517389
Network Card(s):           1 NIC(s) Installed.
                           [01]: Intel(R) PRO/1000 MT Desktop Adapter
                               Connection Name: Ethernet
                               DHCP Enabled:    Yes
                               DHCP Server:     10.0.2.2
                               IP address(es)  [01]: 10.0.2.15
                                                [02]: fe80::6514:b4e9:3670:7024
                                                [03]: fd17:625c:f037:2:4c16:fd50:6831:d4f9
                                                [04]: fd17:625c:f037:2:6514:b4e9:3670:7024
Hyper-V Requirements:      A hypervisor has been detected. Features required for Hyper-V will not be displayed.

```

Figure 2: Virtual Machine System Info

2.3 Verification and Compliance

I verified the hashes of the DFI then proceeded with the investigation. The drive was formatted as follows: it is important to note that the MD5 hash stored is **21975d0f9f2146f3aca1f332ca456bb6** and the SHA1 hash stored is **83472522c6f28debda8862d6339b749dcf392a2e**; this is the changed order of what I was initially informed. There are no bad blocks to report. In order to verify this I used both FTK Image Finder and the built in resource on Autopsy, the proof of which will be shown below.

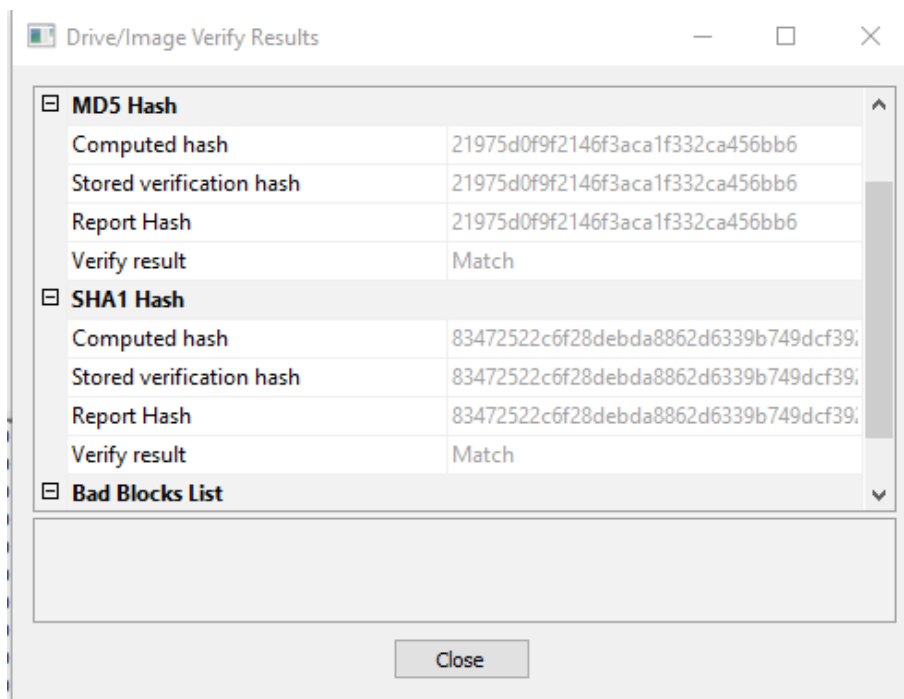


Figure 3: FTK Verification

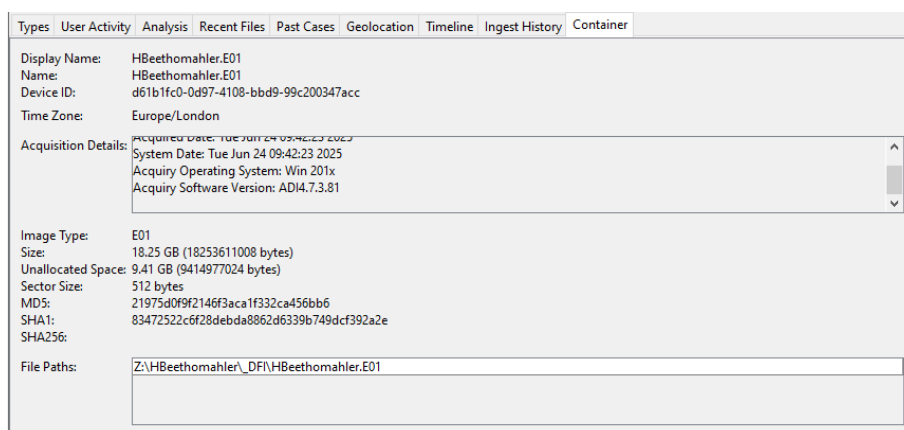


Figure 4: Autopsy Verification

I confirm that, to the best of my knowledge and belief, I have complied with the Code of Practice published by the statutory Forensic Science Regulator 2025 version 2. I have ensured that I have used validated tools and techniques (section 8), upheld data integrity of the case (section 9), legal and ethical compliance (section 10) and accurate representation of the file (section 11) (Government of the United Kingdom, 2025).

3 Medium

3.1 Partition and File System Analysis

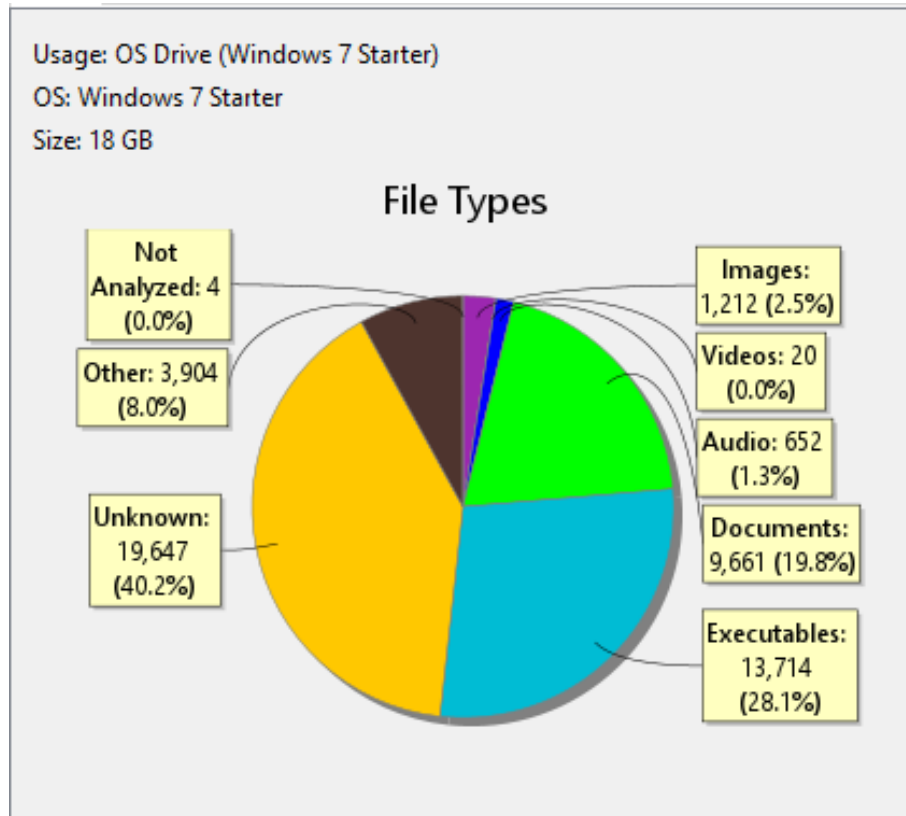


Figure 5: Memory Allocation Chart

The system appears to have a total of 48,778 allocated files, 36 unallocated files, 43,385 slack files and 29,056 directories. The physical disk contains four separate volumes. The disk appears to be populated with NTFS / exFAT (0x07) file system types with the rest of the volumes being unallocated. The implementation of exFAT means that it only uses FAT chains for fragmented files, FAT indices are not cleared after file deletion occurs. This suggests to me that file reconstruction is possible for some deleted files(ExF, 2026). There are limitations to the implementation of exFAT on systems. For example, exFAT is neither able to produce descriptive information for the root directory nor is it able to consistently populate time zone information fields (ExF, 2026).

Name	ID	Starting Sector	Length in Sectors	Description	Flags
vol1 (Unallocated: 0-2047)	1	0	2048	Unallocated	Unallocated
vol2 (NTFS / exFAT (0x07): 2048-206847)	2	2048	204800	NTFS / exFAT (0x07)	Allocated
vol3 (NTFS / exFAT (0x07): 206848-35649535)	3	206848	35442688	NTFS / exFAT (0x07)	Allocated
vol4 (Unallocated: 35649536-35651583)	4	35649536	2048	Unallocated	Unallocated

Table 2: Disk Volume Layout

There appears to be no evidence for unallocated space belonging to "orphaned" data on volumes in the system. Volume 4 is an example of slack space, all data has been over written by 0s, this is unlike volume 1 which has not been completely wiped. Volume 1 contains a few remnants of information that is able to reveal that it has been used in order to write.

3.2 System Configuration and Anomalies

On this system there is evidence for system-reserved areas and potential hidden volumes. I have identified two storage anomalies. For example, volume 1 and volume 4 are potential hidden volumes that are filled with unallocated memory. This suggests that some form of data concealment has taken place on this system.

3.3 Security, Obfuscation and Content Logic

Obfuscation can be achieved through encryption. Encryption is a mathematical function using a secret value—the key—which encodes data so that only users with access to that key can read the information. Information is encrypted and decrypted using a secret key. (Some algorithms use a different key for encryption and decryption). Without the key the information cannot be accessed and is therefore protected from unauthorised or unlawful processing (Information Commissioner’s Office, 2022).

I have carried out a screening for encryption indicators on the items contained within Volume 3. The results returned seven items

with a likely notable score for having unusually high signs of entropy, which I have provided below:

Source Name	File Path
AgCx_SC4.db	/img_HBeethomahler.E01/vol_-vol3/Windows/Prefetch/AgCx_SC4.db
SpeechUXRes.dll	/img_HBeethomahler.E01/vol_-vol3/Windows/System32/Speech/SpeechUX/en-gb/SpeechUXRes.dll
SpeechUXRes.dll	/img_HBeethomahler.E01/vol_-vol3/Windows/System32/Speech/SpeechUX/en-us/SpeechUXRes.dll
SpeechUXRes.dll	/img_HBeethomahler.E01/vol_-vol3/Windows/winsxs/x86_microsoft-windows-s..dlanguage_en-gb.ale_-31bf3856ad364e35_6.1.7600.16385_en-gb_-fbedbf956861a9d/SpeechUXRes.dll
SpeechUXRes.dll	/img_HBeethomahler.E01/vol_-vol3/Windows/winsxs/x86_microsoft-windows-s..dlanguage_en-us.ale_-31bf3856ad364e35_6.1.7600.16385_en-us_-f33773ca928c4cbf/SpeechUXRes.dll
System.Web.dll	/img_HBeethomahler.E01/vol_-vol3/Windows/winsxs/x86_system.web_-b03f5f7f11d50a3a_6.1.7600.16385_none_-cbaefa77d2cfd5e6/System.Web.dll
System.Web.dll	/img_HBeethomahler.E01/vol_-vol3/Windows/winsxs/x86_system.web_-b03f5f7f11d50a3a_6.1.7600.20804_none_-b4d9679fec7e52d7/System.Web.dll

I employed the use of the "Tree View" provided by Autopsy in order to provide a view of the data that is represented on the system. This enabled me to easily traverse the system to find points of interest, it was particularly helpful for flagging notable items for my investigation. An example of this "Tree View" is provided below:

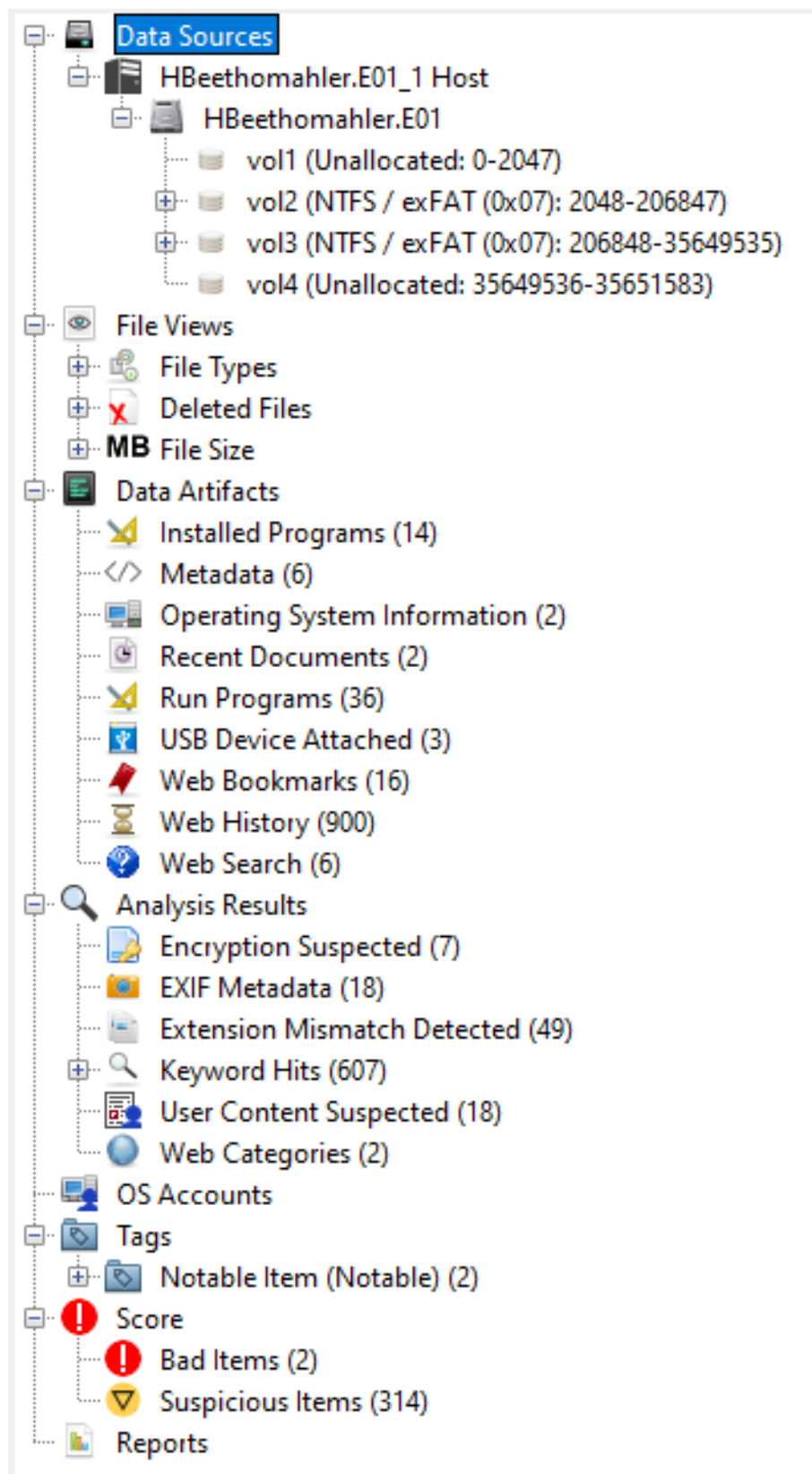


Figure 6: Example of Tree View

4 Operating System

4.1 OS Profiling and Enviornment

The general information for the OS Profiles are stored at the file path **/img HBeethomahler.E01/vol vol3/Windows/System32/co**. By using this file path I was able to establish that the system (BEETHOMAHLER-PC) uses Windows_NT with x86 processor architecture. The operating system used is Windows 7 Starter with the product ID of **55041-946-6744545-85033**. The time zone configured on the system is Pacific Standard Time. Below I have included a table of valuable information that relates to this:

Artifact ID	-9223372036854775779
Date/Time	2023-02-11 05:11:45 GMT
Organization	
Owner	Beethomahler
Path	C:\Windows
Product ID	55041-946-6744545-85033
Program Name	Windows 7 Starter
Source File Path	/img_HBeethomahler.E01/vol_vol3/Windows/System32/config/SOFTWARE

Table 3: OS Information

4.2 User Attribution and Security

Autopsy appears to use 2 different time zones to record activity BST and GMT, GMT is the standard for winter whilst BST is the standard of summer and 1 hour ahead of GMT. Therefore, the recorded times can be amended to 08:08:17 GMT and 04:53:58 GMT. It is worthy to note that there are 9 known user profiles that have been created. Out of the 9 profiles that exist only 2 have shown any login activity: Beethomahler and Administrator. The Beethomahler profile was created in 2023-02-11 05:11:37 GMT and was last used on 2025-06-24 09:08:17 BST with a total of 4 logins. The Administrator profile was created in 2023-02-11 05:11:07 GMT and was last used on 2009-07-14 05:53:58 BST with a total of 1 login. This reveals to me that the Beethomahler was the only true profile that was in use from 2023 to 2025. The user profile administrator does not require a password and has admin privileges; the user profile Beethomahler does not require a password and also holds admin privileges.

Beethomahler is attributed to the Event Log Readers group. This means Beethomahler can read event logs on the machine but does not have the authority to modify system settings. Administrator is the built-in account for system management and has unrestricted access to the computer. This account belongs to the Administrators group, which has complete control over the system. Usersh is part of the Users group. This means Usersh can run many applications but is restricted from making system-wide changes or administrative changes to the system.

Using the SAM database I managed to map the accounts to Security Identifiers (SIDs), the results of which I have provided below in a table:

Name	Login Name	Host	Scope	Realm Name	Creation Time
S-1-5-18	systemprofile	HBeethomahler.E01_1 Host	Local		
S-1-5-80-956008885-3418522649-1831038044-1853292631-2271478464		HBeethomahler.E01_1 Host	Local		
S-1-5-80-2620923248-4247863784-3378508180-2659151310-2535246811		HBeethomahler.E01_1 Host	Local		
S-1-5-19	LocalService	HBeethomahler.E01_1 Host	Local		
S-1-5-20	NetworkService	HBeethomahler.E01_1 Host	Local		
S-1-5-21-4077764705-4164297915-2244365980-1000		HBeethomahler.E01_1 Host	Local		
S-1-5-21-4031847665-3736034123-2700338534-1000	Beethomahler	HBeethomahler.E01_1 Host	Local		2023-02-11 05:11:37 GMT
S-1-5-21-4031847665-3736034123-2700338534-500	Administrator	HBeethomahler.E01_1 Host	Local		2023-02-11 05:11:07 GMT
S-1-5-21-4031847665-3736034123-2700338534-501	Guest	HBeethomahler.E01_1 Host	Local		2023-02-11 05:11:07 GMT

Table 4: User Information Table

4.3 External Connectivity and Connectivity History

BEETHOMAHLER-PC has 3 registered connections to USB devices, which I have listed below. It is unclear what information was shared from or to these devices by the user. The last connected device to Beethomahler PC was the USB tablet (Device ID: 5&18f54cb7&0&1).

Source Name	Date/Time	Device Make	Device Model	Device ID	Data Source
SYSTEM	2025-06-24 17:07:28 BST		ROOT_HUB	4&24d6eb65&0	HBeethomahler.E01
SYSTEM	2025-06-24 17:07:28 BST		ROOT_HUB20	4&6a987e4&0	HBeethomahler.E01
SYSTEM	2025-06-24 17:07:29 BST	VirtualBox	USB Tablet	5&18f54cb7&0&1	HBeethomahler.E01

Table 5: USB Information

The file path used in order to extract information about internet connectivity is `/img_HBeethomahler.E01/vol_vol3/Windows/Sy`. I have extracted saved internet profiles and network history which are able to provide geographic and social context. The internet profiles associated with this machine are Network (**FBDD9DC4-0D6C-4BEE-84A0-0C024002B97D**) and Network 2 (**7C817CAC-0C5A-4C32-9F58-1FB8075A0E19**). I have found an IP address that can be linked to these internet profiles: **192.228.79.201**. It is unclear to me whether this internet connection is Wi-Fi or Ethernet, though I am led to believe that it is an Ethernet connection as it is referred to as a cable. This is a public Wi-fi address. Using WHOSISLLOOKUPIP, I found the relevant information listed

below:

Field	Value
NetRange	192.228.79.0 - 192.228.79.255
CIDR	192.228.79.0/24
NetName	BROOT
NetHandle	NET-192-228-79-0-1
Parent	NET192 (NET-192-0-0-0-0)
NetType	Direct Allocation
OriginAS	(Not provided)
Organization	B.Root-Server-OPS (BROOT)
RegDate	2003-05-01
Updated	2022-08-11
Ref	https://rdap.arin.net/registry/ip/192.228.79.0

Table 6: WHOIS Lookup Table

Field	Type	Value
ProfileName	REG_SZ	Network 2
Description	REG_SZ	Network
Managed	REG_- DWORD	0x00000000 (0)
Category	REG_- DWORD	0x00000001 (1)
DateCreated	REG_BIN	E9 07 06 00 02 00 18 00 00 00 2E 00 20 00 A4 01
NameType	REG_- DWORD	0x00000006 (6)
DateLastConnected	REG_BIN	E9 07 06 00 02 00 18 00 01 00 07 00 2D 00 E0 03
CategoryType	REG_- DWORD	0x00000000 (0)
IconType	REG_- DWORD	0x00000000 (0)

Table 7: Network Connection Table

Field	Type	Value
ProfileGuid	REG_SZ	{FBDD9DC4-0D6C-4BEE-84A0-0C024002B97D}
Description	REG_SZ	Network
Source	REG_DWORD	0x00000008 (8)
DnsSuffix	REG_SZ	cable.virginm.net
FirstNetwork	REG_SZ	Network
DefaultGatewayMac	REG_BINARY	52 54 00 12 35 02

Table 8: Network Connection Table

Field	Type	Value
ProfileGuid	REG_SZ	{7C817CAC-0C5A-4C32-9F58-1FB8075A0E19}
Source	REG_DWORD	0x00000008 (8)
DefaultGatewayMac	REG_BINARY	52 55 0A 00 02 02
DnsSuffix	REG_SZ	cable.virginm.net
Description	REG_SZ	Network 2
FirstNetwork	REG_SZ	Network 2

Table 9: Network Connection Table

Network 2 was last modified at 2025-06-24 07:46:32 GMT +00:00. Network was last modified at 2023-02-10 21:07:12 GMT+00:00. From this information I am able to tell that the Beethomaler PC was used in the UK as Virgin Media is primarily based in the UK, although it is owned by a US based company.

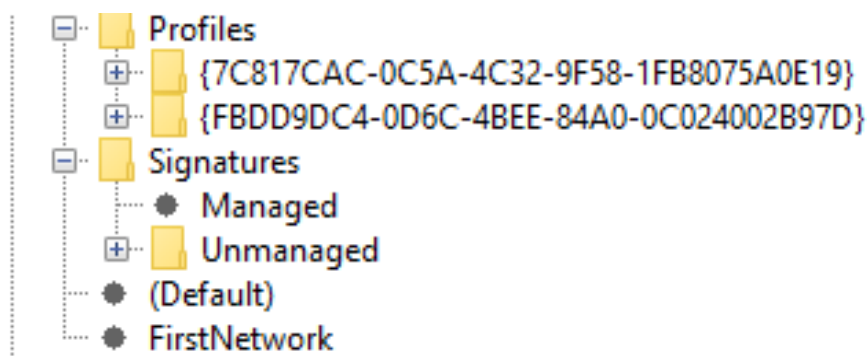


Figure 7: Wi-Fi Profiles in Tree View

5 Conclusion

In conclusion, the analysis of the Beethomaler PC has revealed to me that Beethomaler was operating on a Windows 7 system that had evidence for hidden storage mediums. My investigation of this DFI has followed ACPO to the best of my abilities, such as verification of hashes, ensuring a forensically secure workstation and not working on the original storage media. This ensures the reliability of my report. Overall, I believe my report to be sufficient for the task.

References

- Exfat implementation notes. https://wiki.sleuthkit.org/index.php?title=ExFAT_Implementation_Notes, 2026. SleuthKitWiki. Accessed: 13 February 2026.
- Government of the United Kingdom. Code of practice. Technical report, Government of the United Kingdom, 2025. URL https://assets.publishing.service.gov.uk/media/68401f4be550203c8209cce9/2025_05_20_Code_of_Practice_final.pdf. Accessed: 2026-02-26.
- Information Commissioner's Office. *Encryption Guidance*. Information Commissioner's Office, 2022. URL <https://ico.org.uk/media2/bi0bcdxr/encryption-guidance.pdf>. Version 1.0.2, 17 October 2022.